ACEFA
AUSTRALIA-AOTEAROA CONSORTIUM FOR EPIDEMIC FORECASTING & ANALYTICS

Lecture 2
Epidemic theory 1: typical epidemic dynamics and the susceptible-infectious-removed (SIR) model

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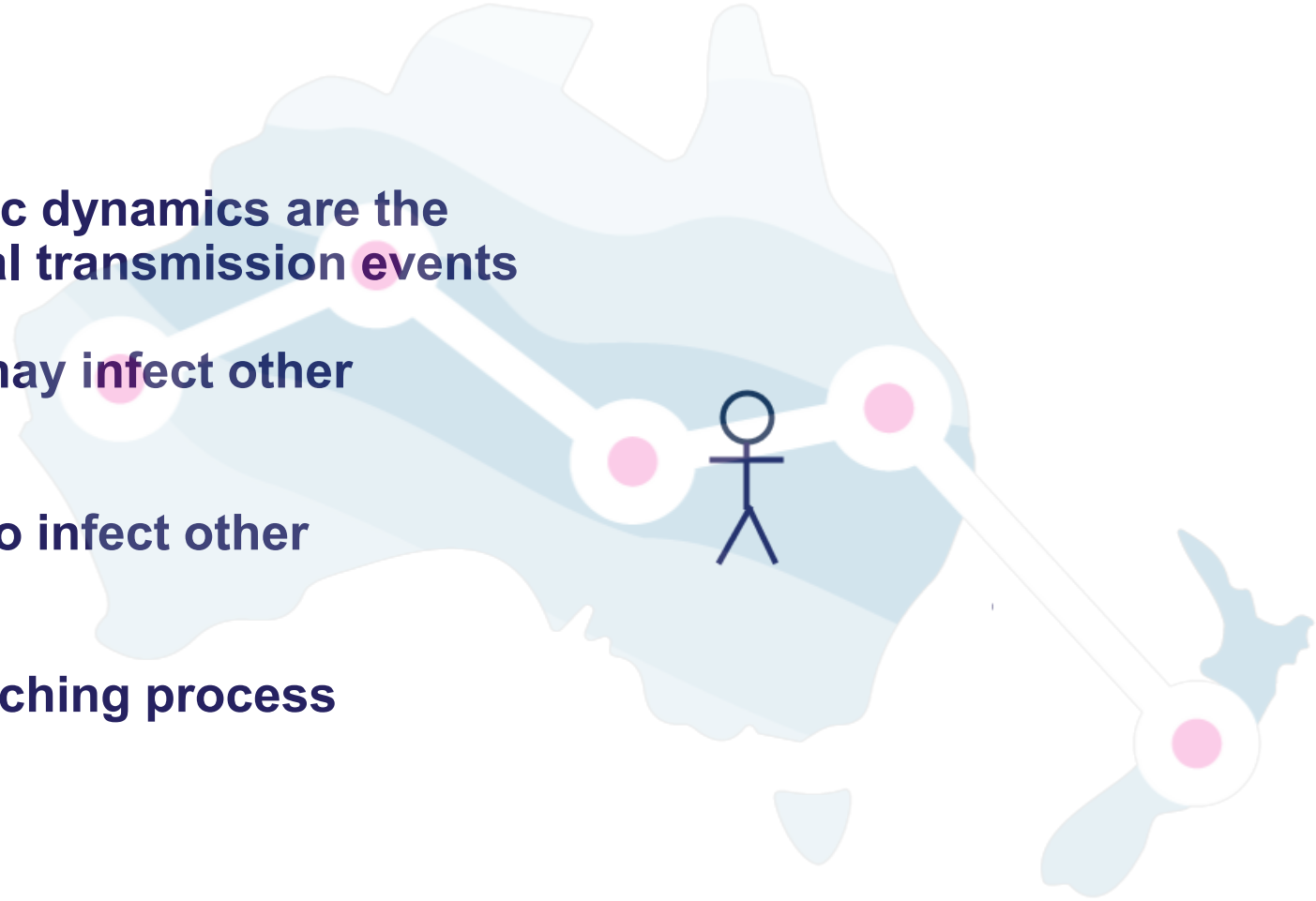
Prepared for the ACEFA EpiAnalytics Course at Australian CDC, Canberra, November 2026
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Typical epidemic dynamics

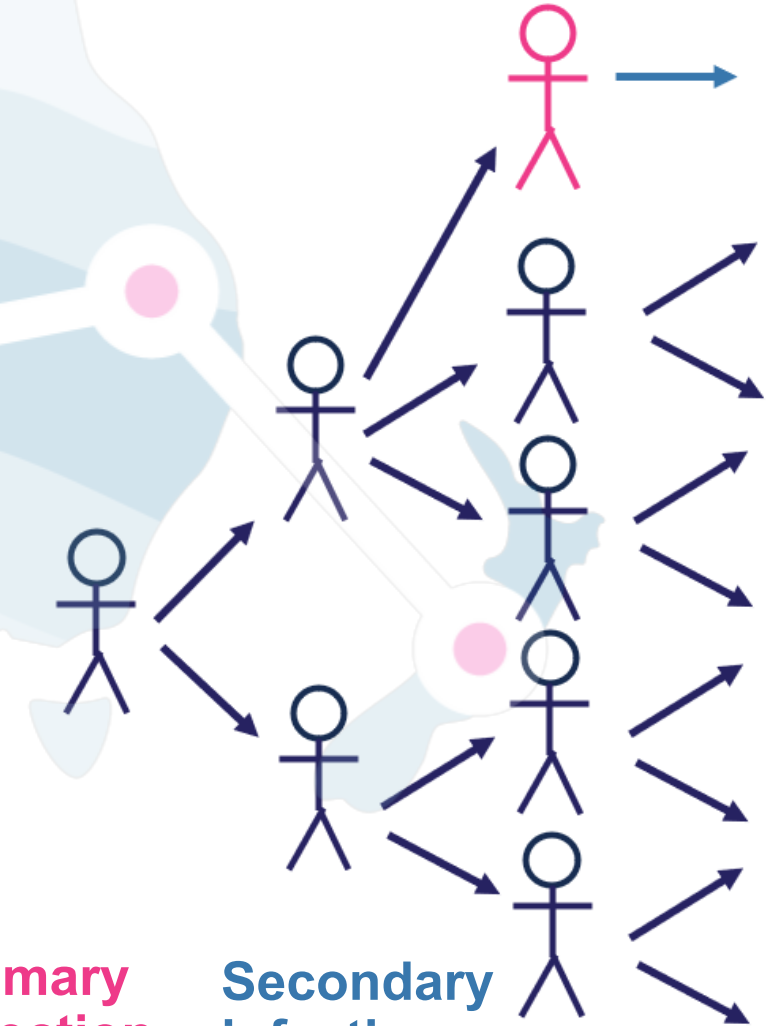
Branching processes

- Fundamentally, epidemic dynamics are the result of many individual transmission events
- An infected individual may infect other individuals
- These may then go on to infect other individuals
- This is known as a branching process



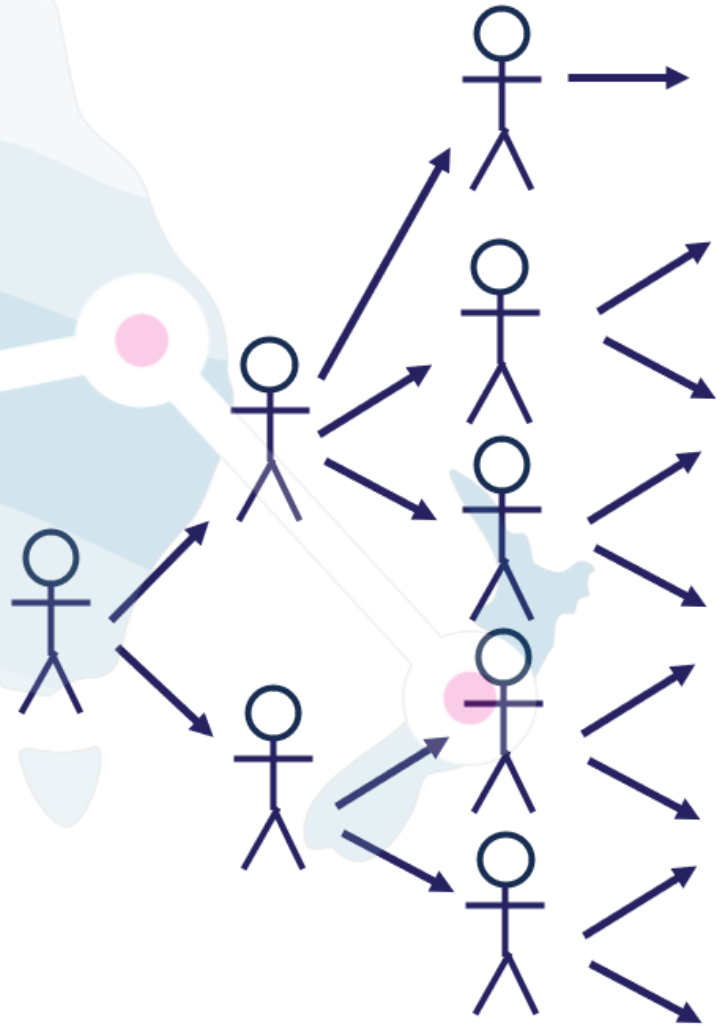
The expected number of infections

- The number of infections caused by an infected individual can vary between individuals
- How many secondary infections will a primary infection cause?
- What factors will contribute?



Group discussion

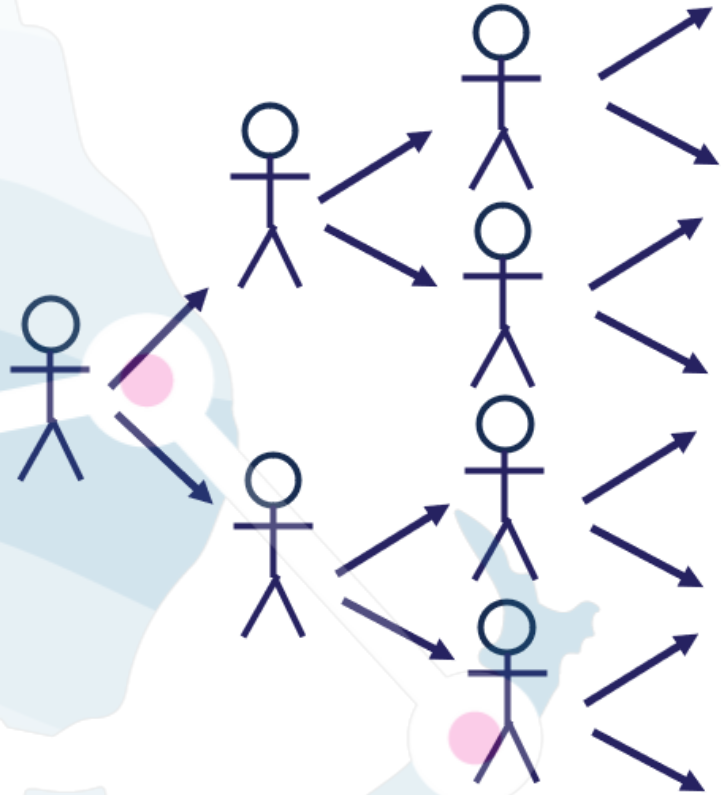
- How many secondary infections will a primary infection cause?
- What factors will contribute?



The expected number of infections

The factors we will consider are:

- The average rate of contact any individual has with other individuals, c
- The probability that contact between an infected individual and a susceptible individual results in infection, p_i
- The probability that contact is with a susceptible individual. If we assume random mixing then equivalent to the proportion of the population that is susceptible, $s(t)$
- The mean duration of infectiousness, μ_i



The expected number of secondary infections is then:

$$c \times p_i \times \mu_i \times s(t)$$

The expected number of infections

c	p_i	μ_i	$s(t)$	Expected number of infections
10	0.1	2	1.0	

The expected number of secondary infections is then:

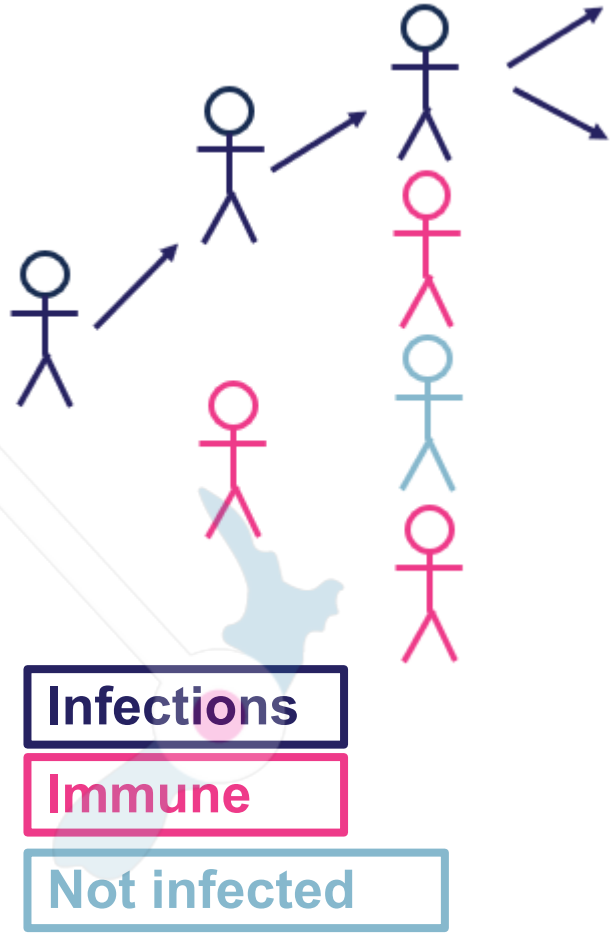
$$c \times p_i \times \mu_i \times s(t)$$

What will happen if the expected number of infections is less than one?

Depletion of susceptibles

The depletion of susceptible individuals will lead to epidemic decline

- **The proportion of the population susceptible, $s(t)$ can change over time**
- **This reduces the expected number of secondary infections**
- **If the expected number of secondary infections <1 then:**
  - **Each primary infection results in fewer secondary infections**

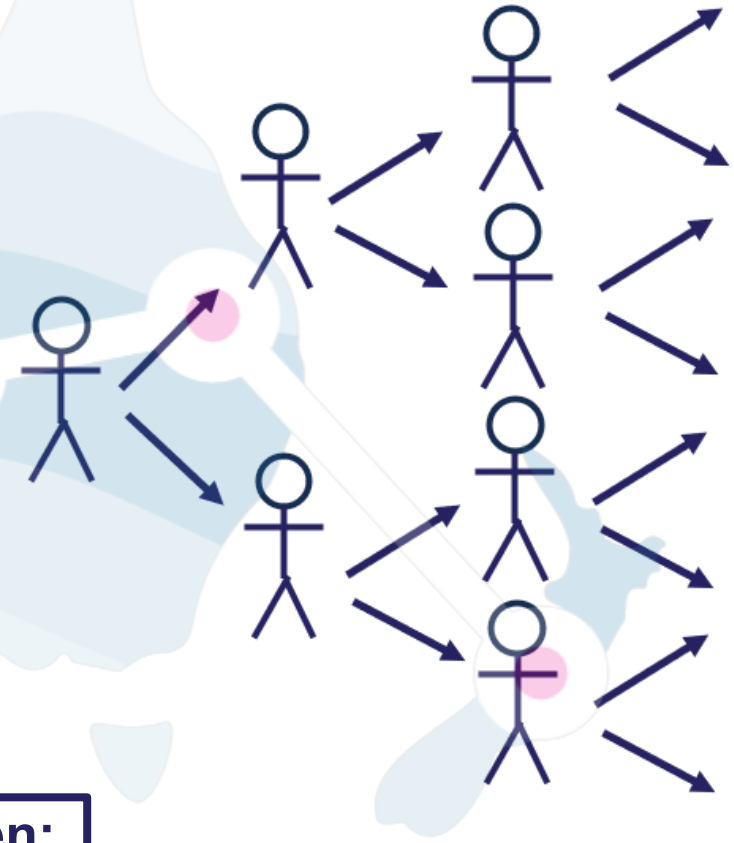


The expected number of secondary infections is then:

$$c \times p_i \times \mu_i \times s(t)$$

Key term: the basic reproduction number, R_0

- The basic reproduction number, R_0, is defined as:
  The expected number of secondary infections generated by one infected individual in a completely susceptible population



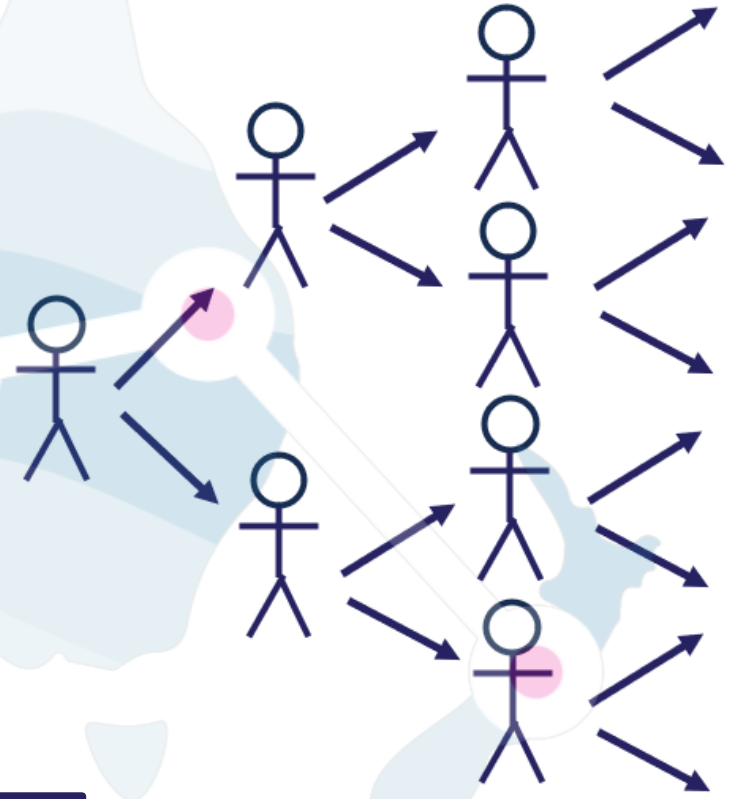
$$R_0 = c \times p_i \times \mu_i$$

The expected number of secondary infections is then:

$$c \times p_i \times \mu_i \times s(t)$$

Key term: the effective reproduction number, R_e

- The effective reproduction number, R_e, is defined as:
  The expected number of secondary infections generated by an infected individual in a population which may not be entirely susceptible and/or in the presence of interventions.



$$R_e = c \times p_i \times \mu_i \times s(t)$$

i.e. the same as the expected number of secondary infections:

$$c \times p_i \times \mu_i \times s(t)$$

The expected number of infections

c	p_i	μ_i	$s(t)$	Expected number of infections
10	0.1	2	1.0	2
20	0.1	2	1.0	4
10	0.2	2	1.0	4
10	0.1	4	1.0	4
10	0.1	2	0.5	1
10	0.1	2	0.25	0.5
5	0.2	2	1	1

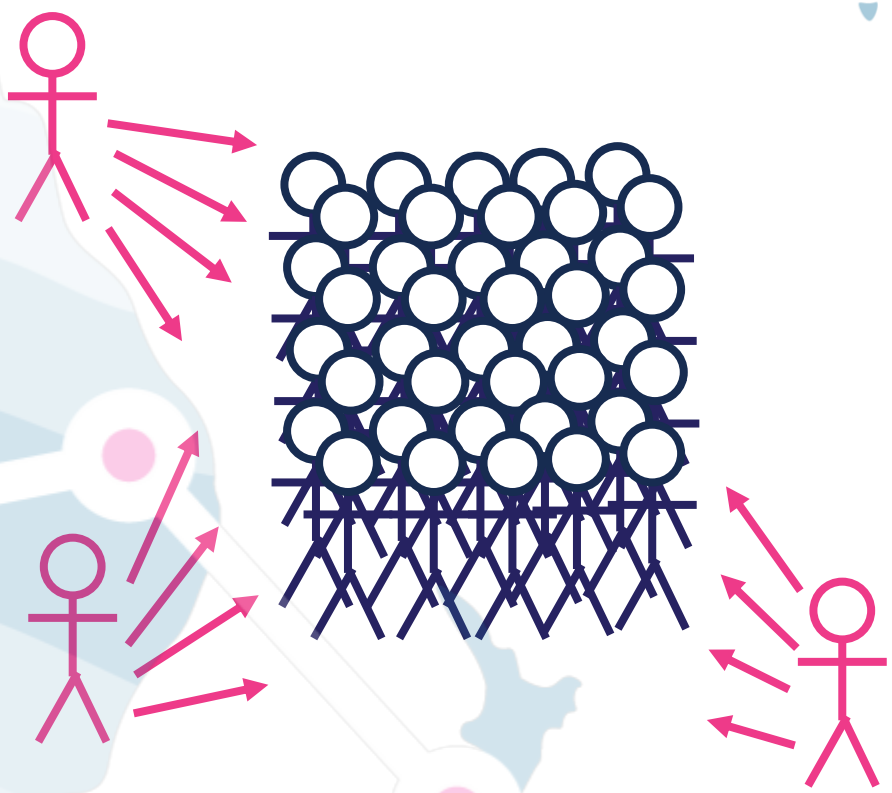
- What is the effective reproduction number in each example?
- What is the basic reproduction number in each example?

Key term: the force of infection

- The force of infection, λ, is defined as:
  The rate at which susceptible individuals acquire an infectious disease

Can use similar reasoning to before:

- The average rate of contact any individual has with other individuals, c
- The probability that contact between an infected individual and a susceptible individual results in infection, p_i
- The probability that contact is with an infected individual. If we assume random mixing then equivalent to the proportion of the population infected, $\mathrm{i}(t)$



The force of infection is then:

$$\lambda = c \times p_i \times i(t)$$

Summary of key terms

- The basic reproduction number, R_0, is defined as:
  The expected number of secondary infections generated by one infected individual in a completely susceptible population

$$R_0 = c \times p_i \times \mu_i$$

- The effective reproduction number, R_e, is defined as:
  The expected number of secondary infections generated by an infected individual in a population which may not be entirely susceptible and/or in the presence of interventions.

$$R_e = c \times p_i \times \mu_i \times s(t)$$

- The force of infection, λ, is defined as:
  The rate at which susceptible individuals acquire an infectious disease

$$\lambda = c \times p_i \times i(t)$$

So how do we measure these parameters?



- These parameters can't always be easily measured
- R_e can often be estimated from epidemic time series (lecture 4) as long as we have an estimate of the generation interval (lecture 3)
- The parameters give us insight into what factors contribute to epidemic growth
- R_e can be reduced by:
  - Reducing contact rates
  - Reducing probability of infection given contact between infected and susceptible individuals
  - Reducing duration of infectiousness
  - Reducing the proportion of the population susceptible
- We can estimate how R_e changes over time to understand the combined impact of interventions

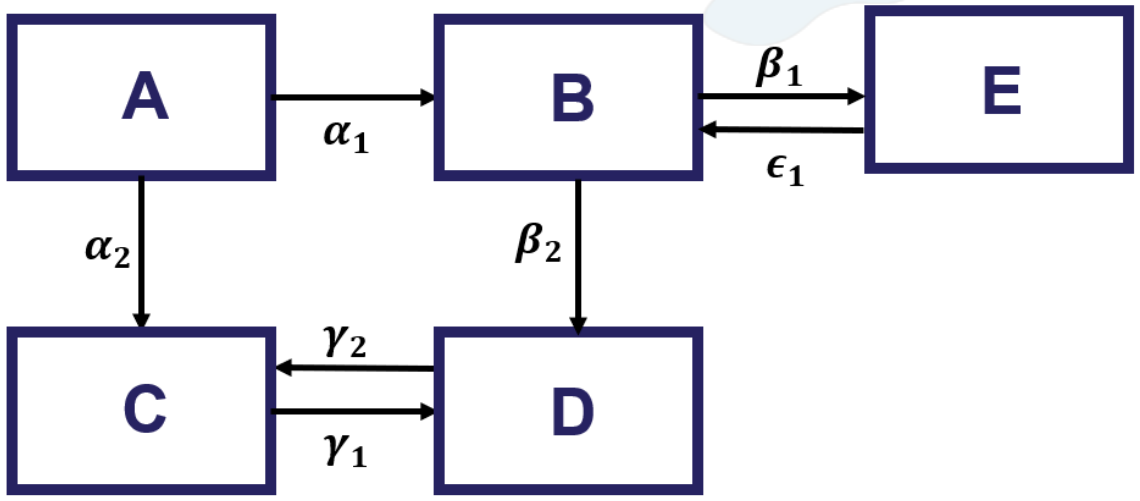
$$R_e = c \times p_i \times \mu_i \times s(t)$$



Compartmental models

Compartmental models

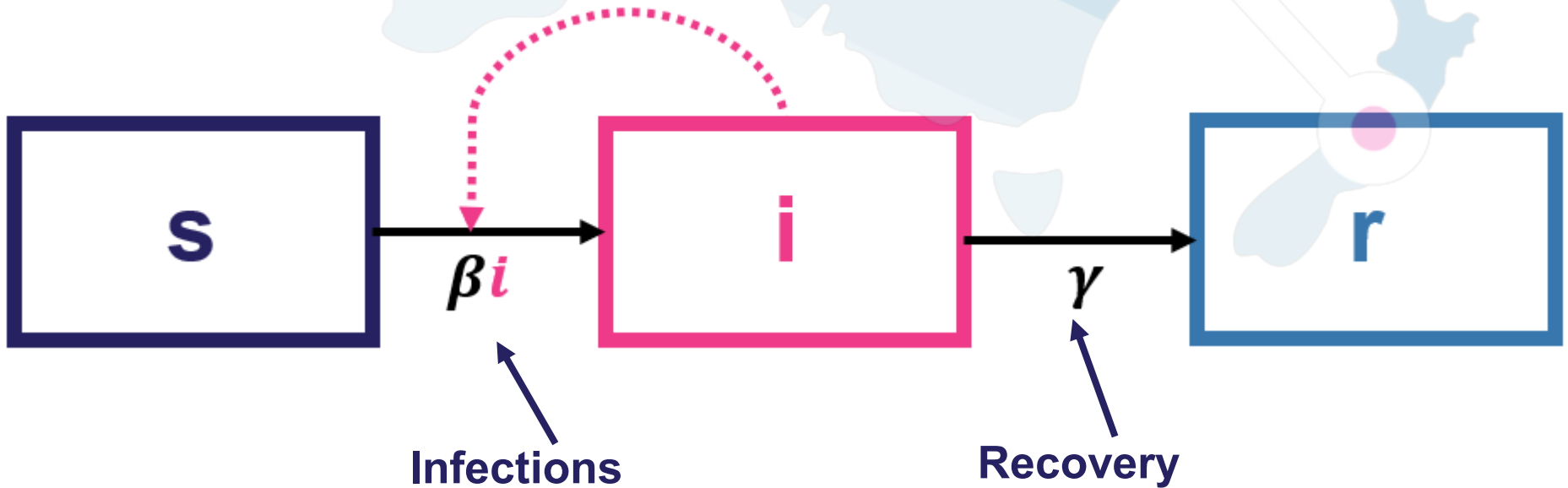
- Compartmental models are a mathematical framework for simulating how populations move between different states or “compartments”
- Compartment models for modelling infectious diseases divide the population into compartments based on their epidemiological status.
- The movement of the population between compartments is defined by a system of equations which define the rate at which individuals flow between each compartment



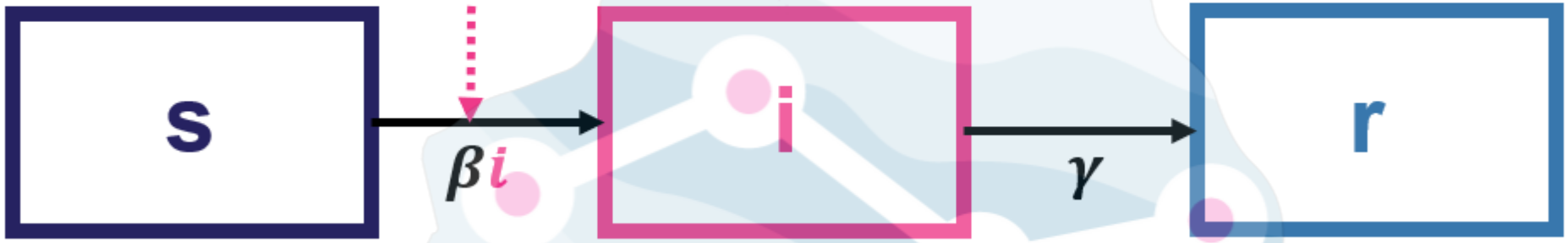
- $\frac{d\,A(t)}{dt} = -\alpha_1 A - \alpha_2 A$
- $\frac{d\,B(t)}{dt} = \alpha_1 A - \beta_1 B - \beta_2 B + \epsilon_1 E$
- Etc.

The SIR model

- The SIR model is one of the simplest compartmental models of infectious diseases
- Each individual is in one of three states:
  - Susceptible (S)
  - Infected and infectious (I)
  - Recovered/removed (R)
- We are interested in how the proportions of the population in each of S, I, and R (s, i, r) change over time



The SIR model



- **Movement between the compartments is governed by the equations:**

- $\frac{d\,s(t)}{dt} = -\beta s(t) i(t)$
- $\frac{di(t)}{dt} = \beta s(t) i(t) - \gamma i(t)$
- $\frac{dr(t)}{dt} = \gamma i(t)$
- $s(t) + i(t) + r(t) = 1$

Parameters

- **Transmissibility rate, β**
- **Recovery rate, γ**

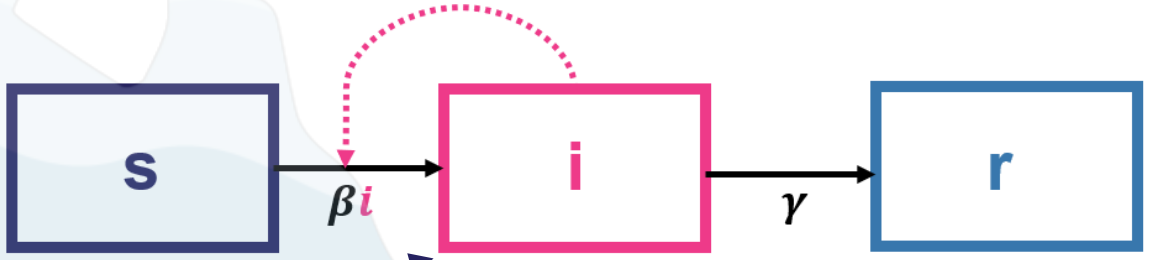
Key term: the prevalence and incidence of infection

- The <u>prevalence of infection</u> is:
  The fraction of the population infected

- In the SIR model the prevalence of infection is $i(t)$

- The <u>incidence of infection</u> is:
  The rate at which new infections occur over a specific time period

- In the SIR model the incidence of infection is $\beta \times i(t) \times s(t)$



- $\frac{d\,s(t)}{dt} = -\beta i(t) s(t)$
- $\frac{di(t)}{dt} = \beta i(t) s(t) - \gamma i(t)$
- $\frac{dr(t)}{dt} = \gamma i(t)$

The SIR model: the transmissibility rate, β

- The force of infection, λ, is defined as:
  The rate at which susceptible individuals acquire an infectious disease

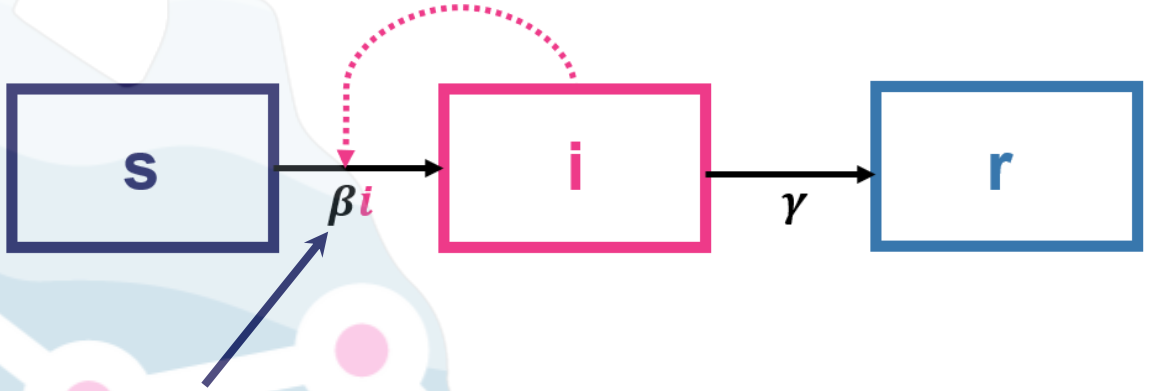
The force of infection is then:

$$\lambda = c \times p_i \times i(t)$$

$$\beta = c \times p_i$$

The transmissibility rate parameter, β is the product of:

- the average contact rate
- the per contact probability of infection



- $\frac{d\,s(t)}{dt} = -\beta i(t) s(t)$
- $\frac{di(t)}{dt} = \beta i(t) s(t) - \gamma i(t)$
- $\frac{dr(t)}{dt} = \gamma i(t)$

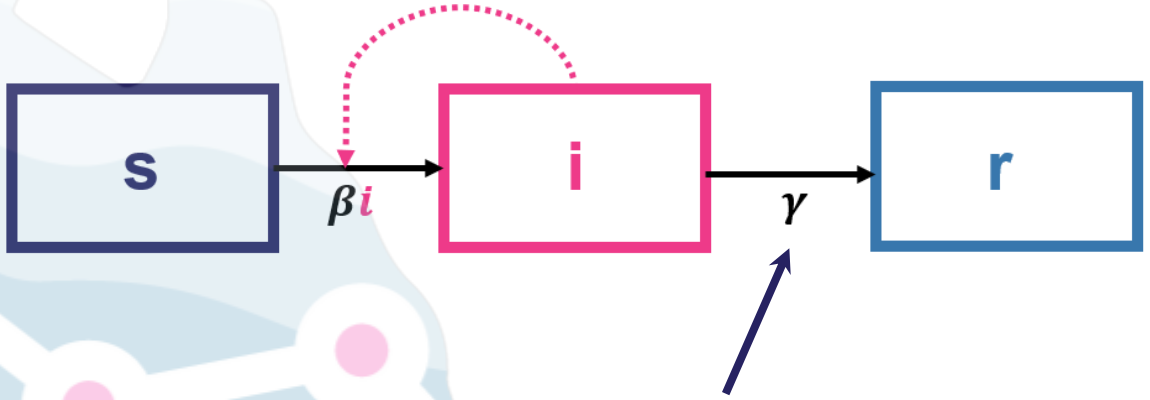
The SIR model: the recovery rate, γ

- The recovery rate, γ, is the rate at which infected individuals recover.
- The average time it takes for an infected individual to recover is given by:

$$\frac{1}{\gamma}$$

- This is equivalent to the mean duration of infectiousness, μ_i.

$$\gamma = \frac{1}{\mu_i}$$



- $\frac{d\,s(t)}{dt} = -\beta i(t) s(t)$
- $\frac{di(t)}{dt} = \beta i(t) s(t) - \gamma i(t)$
- $\frac{dr(t)}{dt} = \gamma i(t)$

Key terms

$$\gamma = \frac{1}{\mu_i}$$

$$\beta = c \times p_i$$

- The basic reproduction number, R_0:

$$R_0 = c \times p_i \times \mu_i \longrightarrow R_0 = \frac{\beta}{\gamma}$$

- The effective reproduction number, R_e:

$$R_e = c \times p_i \times \mu_i \times s(t) \longrightarrow R_e = \frac{\beta}{\gamma} \times s(t)$$

- The force of infection, λ:

$$\lambda = c \times p_i \times i(t) \longrightarrow \lambda = \beta \times i(t)$$

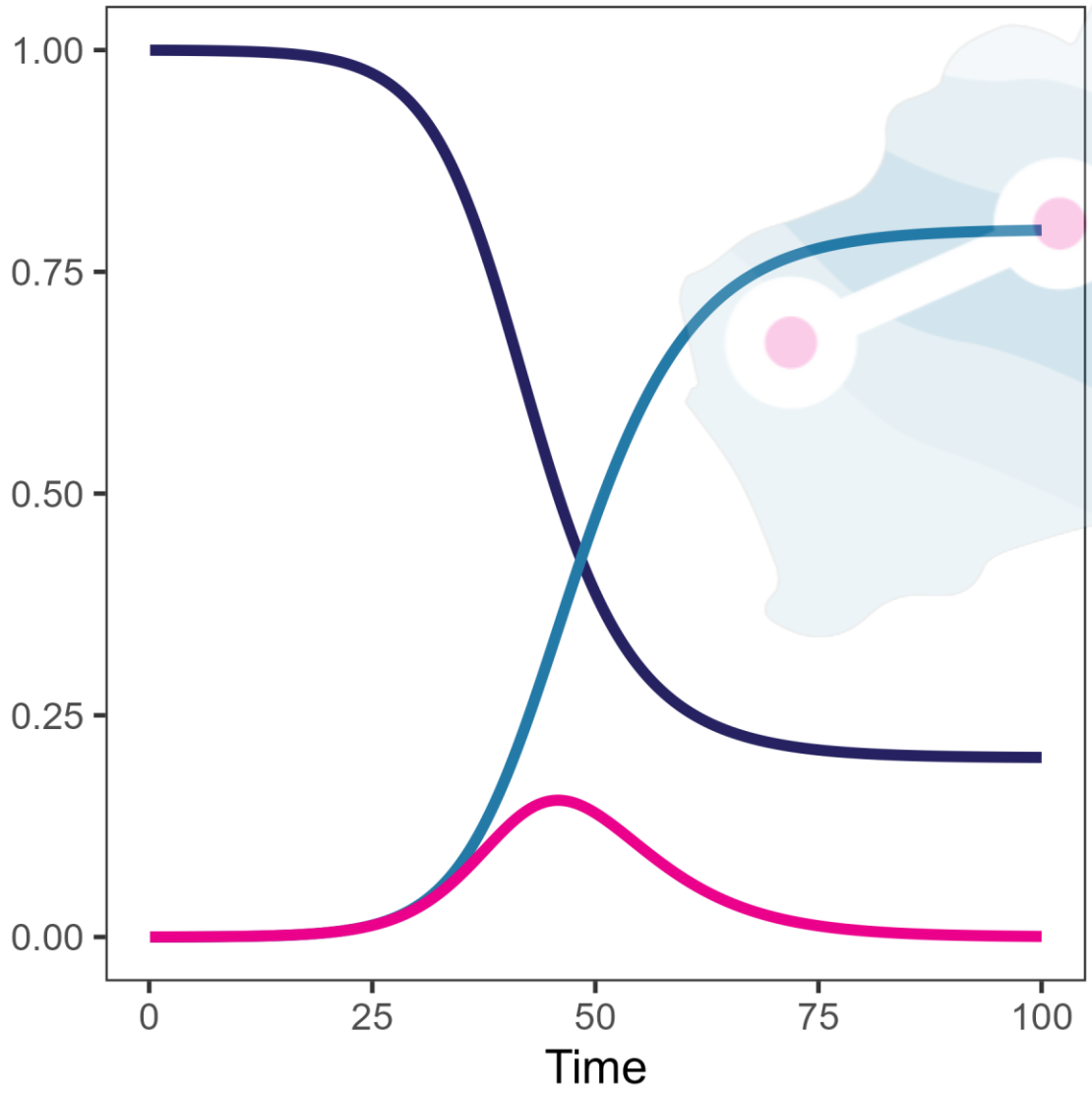
The SIR model: reproduction numbers

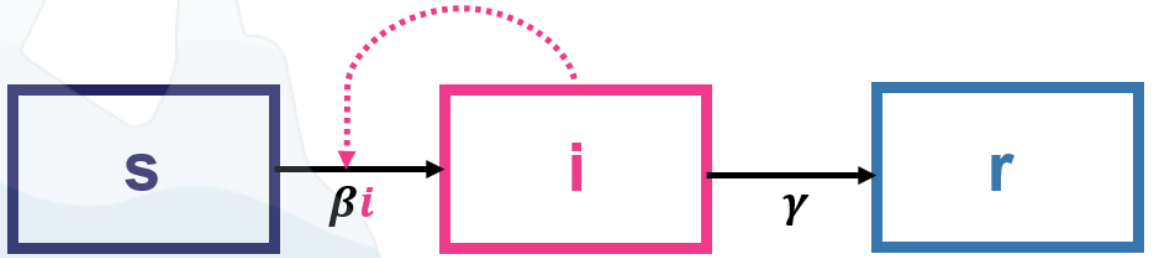
β	γ	$s(t)$	R_0	R_e
1	0.5	1		

$$R_0 = \frac{\beta}{\gamma}$$

$$R_e = \frac{\beta}{\gamma} \times s(t)$$

Group discussion





$$R_0 = \frac{\beta}{\gamma}$$

- $\frac{d\,s(t)}{dt} = -\beta i(t) s(t)$
- $\frac{di(t)}{dt} = \beta i(t) s(t) - \gamma i(t)$
- $\frac{dr(t)}{dt} = \gamma i(t)$

The SIR model: epidemic growth threshold

We can rewrite the second equation as:

$$\frac{di(t)}{dt} = (R_0 s(t) - 1) \times \gamma \times i(t)$$

- Infections increase when:
  $$R_0 \times s(t) > 1$$
- Infections decrease when:
  $$R_0 \times s(t) < 1$$

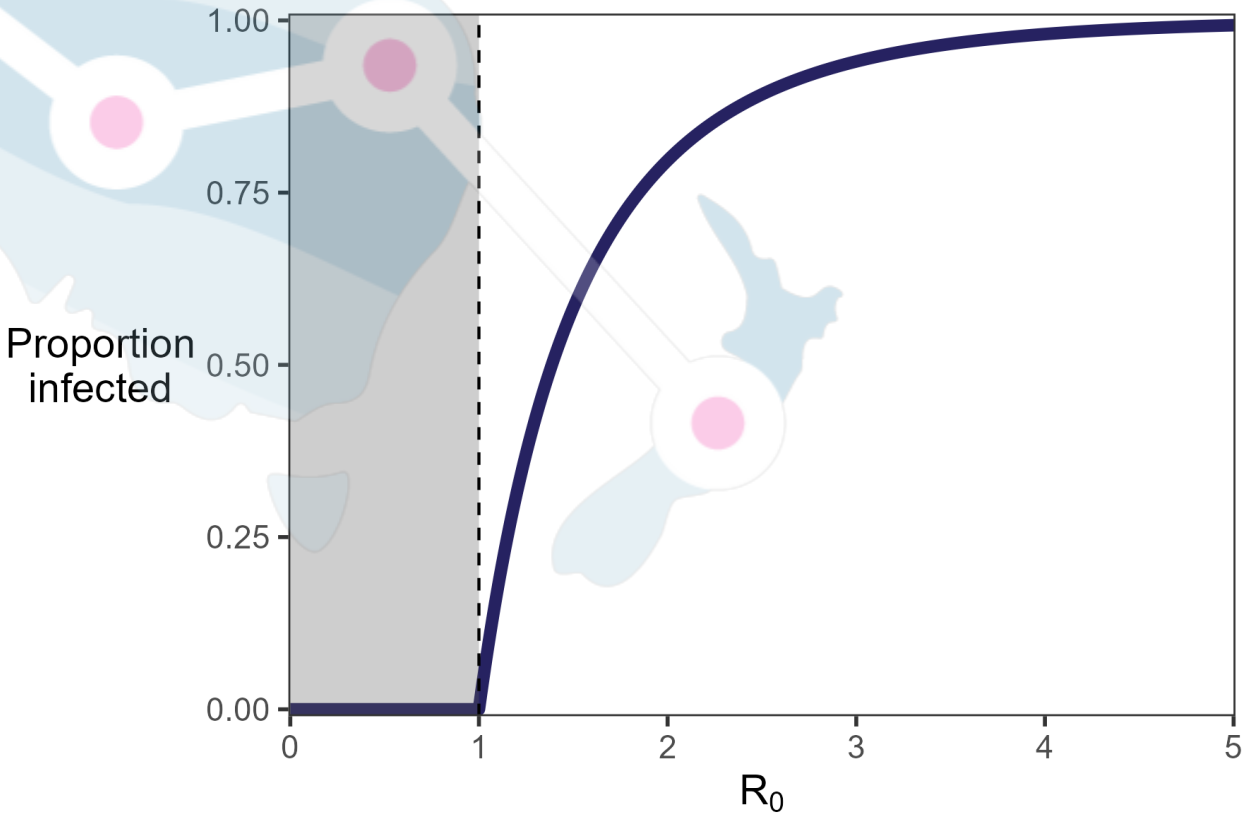
In an entirely susceptible population, i.e. $s(t) = 1$.

$R_0 = 1$ is the threshold for epidemic growth in an entirely susceptible population, i.e.

An epidemic can only occur if $R_0 > 1$

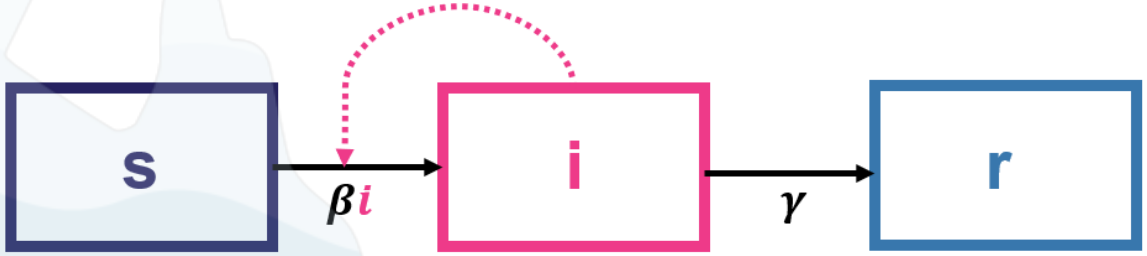
The larger the value of R_0 the bigger the epidemic





The SIR model: herd immunity threshold

- Infections increase when:
  $R_0 \times s(t) = R_e(t) > 1$
- Infections decrease when:
  $R_0 \times s(t) = R_e(t) < 1$

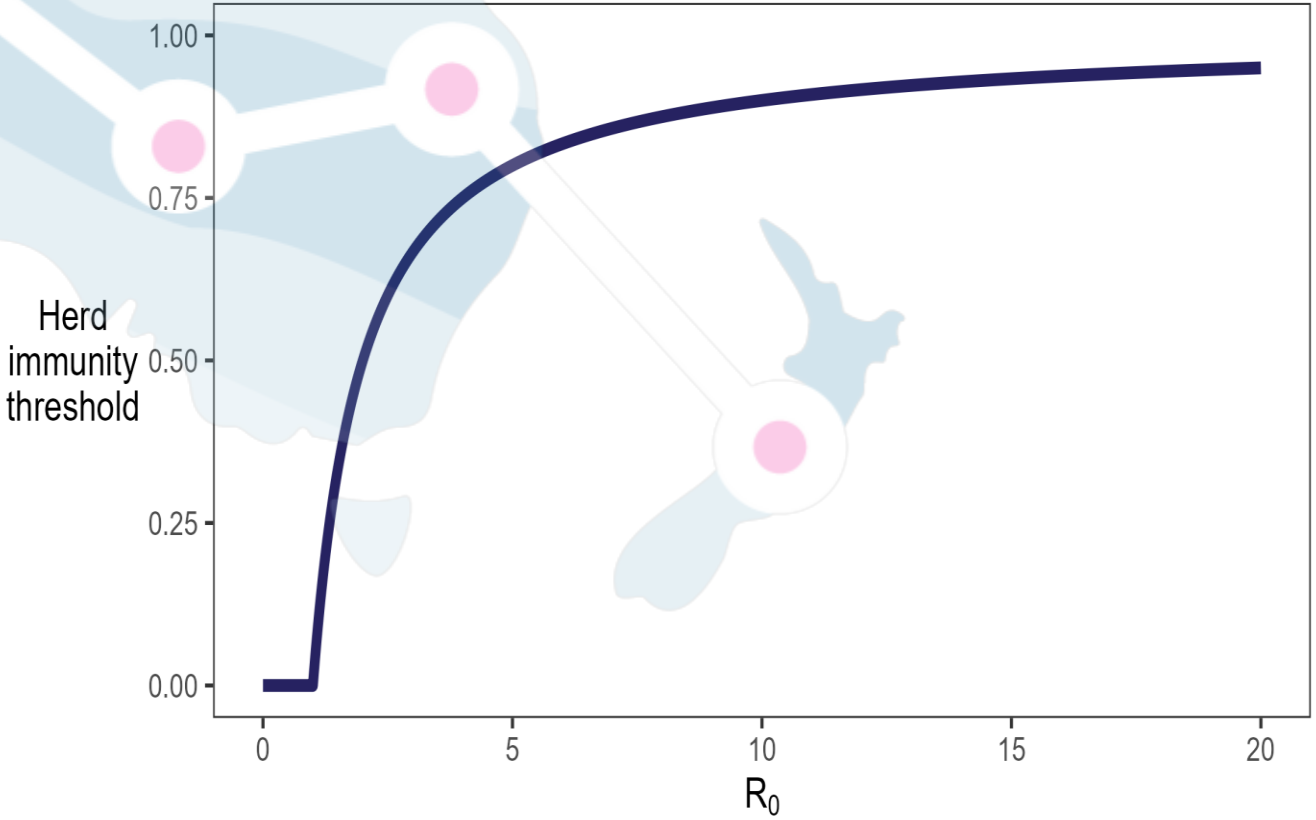


$R_e = 1$ is the threshold for epidemic growth, i.e.

In a population with some immunity an epidemic can only occur if $R_e > 1$

- When $s(t) = \frac{1}{R_0}$: $R_e = 1$

The herd immunity threshold immunity is given by:
$1 - \frac{1}{R_0}$



The SIR model: herd immunity threshold

In a population with some immunity an epidemic can only occur if $R_e > 1$

- **When $s(t) = \frac{1}{R_0} : R_e = 1$**

The herd immunity threshold immunity is given by:
$1 - \frac{1}{R_0}$

Pathogen	R_0	Herd immunity threshold
Influenza	~2	
SARS-CoV-2	~4	
Polio	~7	
Chickenpox	~10	
Measles (lower)	~15	
Measles (upper)	~20	

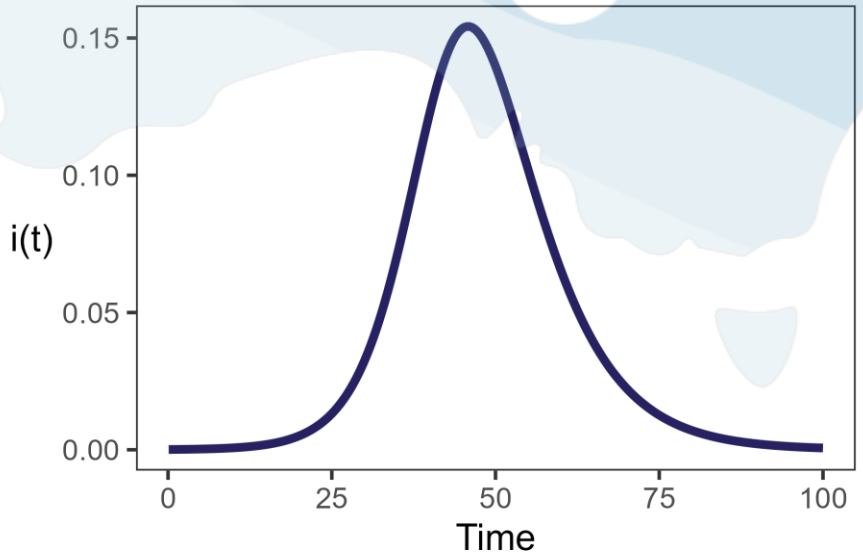
Group discussion

The SIR model is one of the simplest models of infectious disease dynamics.

- **What assumptions does the SIR model make?**

- **What does the SIR model not take into account that might be important for understanding epidemic dynamics?**





- $\frac{d\,s(t)}{dt} = -\beta i(t)s(t)$
- $\frac{di(t)}{dt} = \beta i(t)s(t) - \gamma i(t)$
- $\frac{dr(t)}{dt} = \gamma i(t)$

Some assumptions of the SIR model

- Assumes an individual is infectious as soon as they are infected
- Assumes all individuals have the same rate of contact
- Assumes all individuals are perfectly mixed
- Assumes that the transmissibility rate, β, is constant over time



- $\frac{d\,s(t)}{dt} = -\beta i(t)s(t)$
- $\frac{di(t)}{dt} = \beta i(t)s(t) - \gamma i(t)$
- $\frac{dr(t)}{dt} = \gamma i(t)$

These will be covered in more detail in later lectures

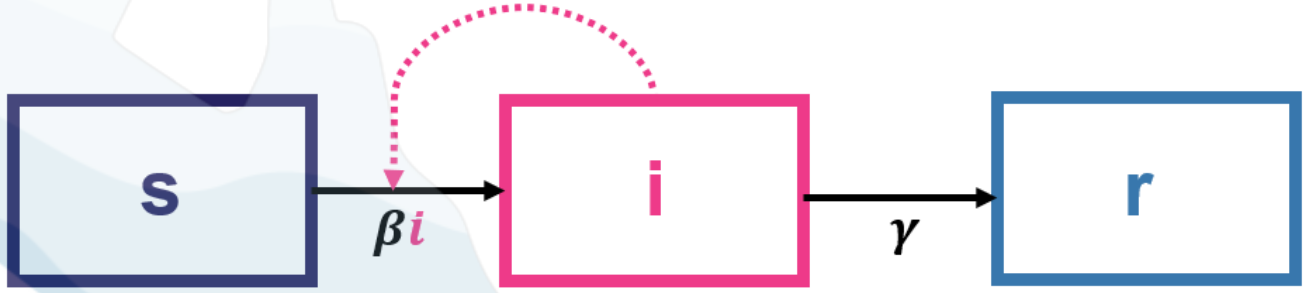
Models should have an appropriate level of complexity



- Models range from simple (e.g. grouping individuals into compartments) or highly complex (e.g. simulating the behaviour of all individuals)
- The complexity of a model depends on the specific research question/purpose, existing knowledge of the system, available data, and computational resources
- Model complexity is not an indicator of its utility
- Often a trade-off between reality (complexity, specificity) and understanding (simplicity, generalisability)
- More complex models require more detailed data for parameterization
- Model "non-identifiability" – multiple different sets of parameter values can produce the same or very similar outputs.

Simulating compartmental models

Compartmental models can be implemented deterministically or stochastically



Deterministic implementation

- There is a predetermined relationship between model inputs and structure and model outputs
- Model will return the same output if input parameters are the same
- Flow between compartments is set by the rates

Stochastic implementation

- Includes inherent randomness in the epidemic dynamics
- Model will return different outputs for the same input parameters
  - Need to run the model many times to capture probability of different outcomes
- Transitions of individuals between compartments occur with probabilities (determined by rates)

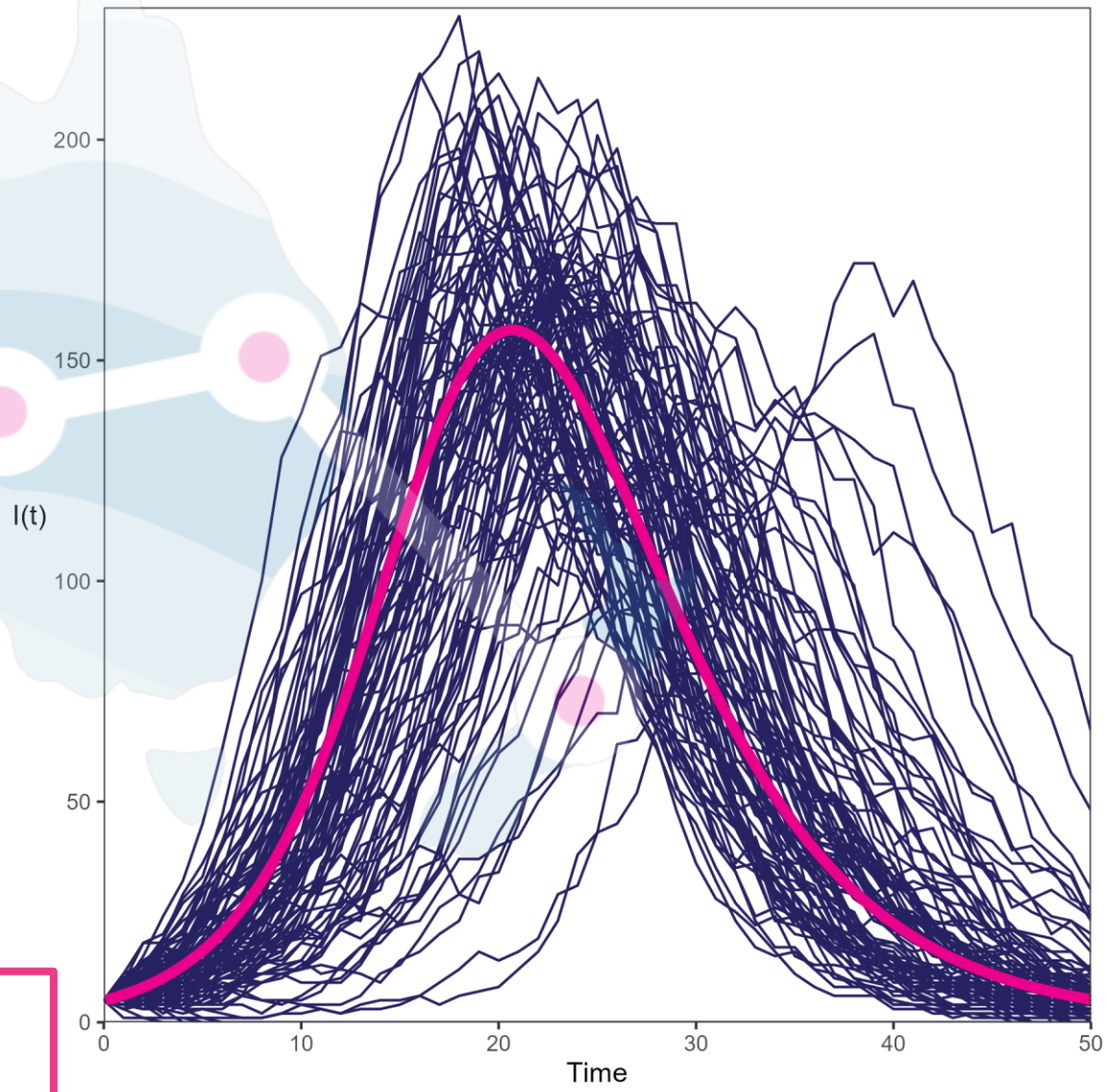
Simulating compartmental models

Compartmental models can be implemented deterministically or stochastically

- **Deterministic models are best suited to modelling epidemic dynamics in large populations**
- **Stochastic models are more useful in small populations or when infection numbers are low (when capturing chance events are important)**

In today's practical we will:

- **Explore the dynamics of the SIR model**
- **Compare stochastic simulations vs a deterministic simulation**



Stochastic simulations

Deterministic solution

References

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